MasterHub IoT Bugfix Report

Noushith project | 5 October 2026

Source workbook: MasterHub_IoT_Noushith_SonarQube_Complete_Analysis.xlsx

Source changes address all 16 BUG findings and all 3 VULNERABILITY findings in the current SonarQube scan scope. Validation completed with 215 passing Python tests and 14 passing JavaScript tests. A fresh SonarQube scan is required before marking these findings closed.

Scope and reconciliation

The supplied workbook contains 4,805 findings across 399 files. Its scan included browser runtime files and historical backups. The existing sonar-project.properties already excludes those directories and generated artifacts. These exclusions were preserved; excluded findings were not repaired or counted as fixed.

Finding type	Workbook	Current scope	Already excluded
Bugs	304	16	288
Vulnerabilities	6	3	3
Code smells	4495	621	3874
Total	4805	640	4165

Outcome

Implemented source corrections for 19 bug and vulnerability findings, plus 6 related dashboard code smells. The remaining 615 in-scope code smells were not addressed in this pass. The 288 excluded bugs comprise 286 browser/runtime findings and 2 historical-backup findings.

Deployment change

Running python app.py now binds to 127.0.0.1 by default. For mobile, MQTT-associated HTTP clients, or other remote clients that access this Flask server, set MASTERHUB_HOST to the PC's LAN interface address before startup. The Flask CLI and external WSGI servers retain their own bind configuration. Debug mode remains disabled.

The two dashboards now use their existing local/system font fallbacks instead of loading Google Fonts stylesheets. Typography may differ on machines without Inter or JetBrains Mono installed.

Dashboard fixes

Unique status identifiers

Removed the hidden duplicate connection matrix from dashboard.html. The six affected identifiers now occur only once, preventing ambiguous DOM lookups and duplicate accessibility content. The visible sidebar retains its original IDs and event handlers.

Keyboard controls

Converted the three dismiss backdrops, three connection rows, BCI status pill, and theme toggle into native buttons. Enter and Space use native activation. Removed the theme toggle's custom keyboard handler to prevent double activation. Added visible focus outlines and preserved full-width connection rows and full-screen backdrop geometry. Closed backdrops are hidden from keyboard focus.

Live sidebar status

The retained sidebar labels previously remained at CHECKING. MQTT, USB, BCI, system health, and FSM mode now update from the existing polling requests. Failed requests replace stale values with Unavailable. This additional functional correction is covered by a JavaScript regression test and is not counted as an extra workbook finding.

Related accessibility cleanup

Removed viewport restrictions that disabled zoom. Native controls also address three non-native-interaction findings and two button-role findings. Existing close controls, Escape handling, IDs, and click actions are preserved.

Backend and diagram fixes

Explicit server interface

app.py reads MASTERHUB_HOST with a localhost fallback, including blank or whitespace-only values. .env.example documents the setting and the startup log reports the selected interface. This removes the unconditional all-interface default flagged by python:S8392.

Device selection

core/devices.py now assigns the normalized target once while holding the existing lock. Registered targets and explicit targets that have not yet been discovered remain valid. This removes identical branches without changing routing semantics.

Diagram arrow rendering

The robotics diagram helper always supplies arc3 with the requested radius. Radius zero already represents a straight connector, so the float equality comparison and alternate branch are unnecessary.

External stylesheets

Removed the remote Google Fonts stylesheet from dashboard.html and cortex_classic.html. Both pages retain their existing font-family fallback stacks, eliminating the two reported unverified external stylesheet loads.

Validation and remaining work

Validation	Result
Full Python regression suite	215 passed in 6.88 seconds
JavaScript regression suites	14 passed
New Python regression cases	10 cases included in the full suite
New JavaScript regression case	Success and failure of sidebar status refresh
SonarQube rescan	Not run; scanner command unavailable
Physical device and browser checks	Live hardware and interactive browser checks not performed

Python tests cover rendered dashboard ID uniqueness and native controls, explicit target selection, default and overridden server binding, and removal of remote stylesheet links. Existing tests exercise API routing, device categories, Cortex behavior, registration, workflow, and origin protection. JavaScript tests cover status updates, safe text rendering, and temporal navigation.

The Python run emitted three warnings: two helper classes with constructors were not collected as test classes, and the modernization suite emitted its existing initialization warning. There were no test failures.

Reproduction commands

python -m pytest tests -q --basetemp=deliverables/sonarqube_fix/pytest-full

node --test tests/test_dashboard_security.cjs tests/test_dashboard_temporal.cjs

Required follow-up

Run the configured SonarQube project again and reconcile the 19 bug/vulnerability findings and 6 related code smells against the new results. These are source-level remediations, not server-confirmed closures. No post-fix quality-gate status or zero-bug claim is made for the complete workbook.

Review the 615 remaining in-scope code smells separately, especially complexity and exception handling. Third-party browser findings should be handled through vendor updates or a separate dependency review; backup findings remain historical. The current scan exclusions were not expanded to improve the totals.

Before deployment, confirm the chosen LAN bind address where remote access is needed, exercise keyboard navigation and dashboard layout in the target browser, and verify MQTT, serial, and headset flows with the actual devices.

Change and evidence records

Eight application/configuration files changed: app.py, .env.example, core/devices.py, docs/generate_robotics_iot_diagrams.py, templates/dashboard.html, templates/cortex_classic.html, static/masterhub.css, and static/masterhub.js. Added tests/test_sonarqube_fixes.py and extended tests/test_dashboard_security.cjs.

The sonarqube_fix deliverable folder retains original copies of the eight files, a unified patch, test outputs, and finding_disposition.csv with all 4,805 workbook rows classified. No Git metadata was present in the workspace.

Bug and vulnerability register

All entries below are implemented in source and await SonarQube confirmation. Lines refer to the supplied workbook, because editing shifts source line numbers. Dashboard entries refer to templates/dashboard.html.

Source	Rule	Correction
app.py line 82	python:S8392	Default bind to localhost
core/devices.py line 128	python:S3923	One normalized assignment
Robotics diagram generator line 36	python:S1244	Always use arc3 radius
Cortex Classic line 7	Web:S5725	Remove remote font stylesheet
Dashboard line 7	Web:S5725	Remove remote font stylesheet
Dashboard line 26	Mouse keyboard equivalent	Native keyboard button
Dashboard line 29	Mouse keyboard equivalent	Native keyboard button
Dashboard line 49	Mouse keyboard equivalent	Native keyboard button
Dashboard line 53	Mouse keyboard equivalent	Native keyboard button
Dashboard line 61	Mouse keyboard equivalent	Native keyboard button
Dashboard line 173	Web:S7930	Remove duplicate sidebar-status-matrix
Dashboard line 175	Web:S7930	Remove duplicate sideMqttStatus
Dashboard line 176	Web:S7930	Remove duplicate sideUartStatus
Dashboard line 177	Web:S7930	Remove duplicate sideBciStatus
Dashboard line 178	Web:S7930	Remove duplicate sideHealthStatus
Dashboard line 179	Web:S7930	Remove duplicate sideActiveFsmMode
Dashboard line 184	Mouse keyboard equivalent	Native keyboard button
Dashboard line 187	Mouse keyboard equivalent	Native keyboard button
Dashboard line 493	Mouse keyboard equivalent	Native keyboard button